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TRANSFORMATION OF AGING EHV SUBSTATION ASSETS THROUGH AUTOMATION AND DIGITAL TECHNOLOGIES FOR IMPROVED SYSTEM RELIABILITY

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Introduction

TRANSFORMATION OF AGING EHV SUBSTATION ASSETS THROUGH AUTOMATION & DIGITAL TECHNOLOGIES FOR IMPROVED SYSTEM RELIABILITY

- **Extra High Voltage (EHV) substations is the backbone of modern power transmission networks**, ensuring reliable bulk power transfer across regions and countries.
- At the same time**, growing load demand, renewable energy integration, grid comp**lexity and reliability expectations are placing unprecedented stress on legacy infrastructure.
- **Utilities are increasingly adopting automation and digital technologies** to modernize aging assets while extending their operational life.
- Risk-Based Digital Transformation of Aging EHV Substations , **need for predictive analytics and centralized asset intelligence**
- Pilot initiative by MSETCL take and established **Maharashtra Transmission Asset Management Centre (MTAMC) at Airoli Navi Mumbai**

MAHATRANSCO
Maharashtra State Electricity Transmission Co. Ltd

MTAMC
Maharashtra Transmission Asset Management Centre.

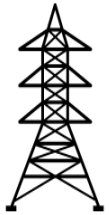
MSETCL Assets and Key challenges

TRANSFORMATION OF AGING EHV SUBSTATION ASSETS THROUGH AUTOMATION & DIGITAL TECHNOLOGIES FOR IMPROVED SYSTEM RELIABILITY

Transmission lines

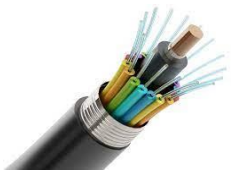
51,518 CKM

Towers



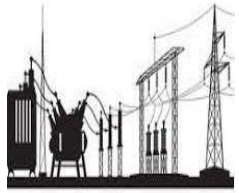
@1,00,000 Nos.

OPGW Network



3,300 KM

EHV Substations



742 Nos.

Transmission Capacity

1,38,598 MVA

Manpower

15,339 Nos.

Key challenges in aging substations

- 30-40 years old infrastructure & Limited visibility of equipment health
- Availability of data in silos and need of Centralized situational awareness
- Decentralised and slow fault diagnosis.
- Digitization of legacy system and cybersecurity implementation.
- Complex integration of legacy infrastructure with modern digital systems

Maharashtra Transmission Asset Management Centre (MTAMC) of MSETCL

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- MSETCL has established the MTAMC (Maharashtra Transmission Asset Monitoring Centre) to enable centralized, real-time asset monitoring and analytics.
- A pilot project has been launched integrating nine 400 kV substations into the centralized monitoring framework.
- Initiative aims to enhance system reliability, minimize unplanned outages, optimize maintenance strategies, and accelerate Digital Grid Transformation.

Key systems implemented under MTAMC

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SCADA - Centralized Remote Monitoring & Operation of EHV Substation

RAS - Proper co-ordination of protection relay Settings

AFAS - Improved system reliability through faster automated analysis of occurrences and even

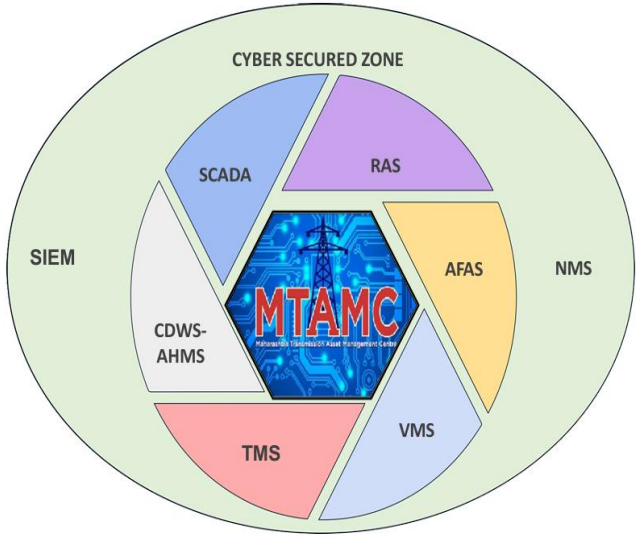
CDWS - Creating repository of asset data to have a single source of truth for power system Analytics

AHMS & TMS - Improved asset life expectancy by having meaningful insights into asset health indicators

VMS - Real-time video surveillance for off-site monitoring of critical assets

SIEM - Security Information and Event Management.

NMS - Network Management System.



Major achievements under MTAMC

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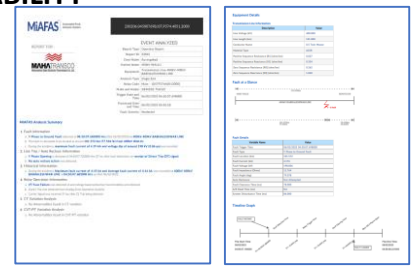
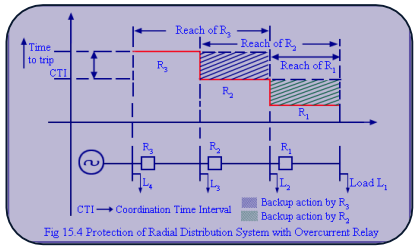


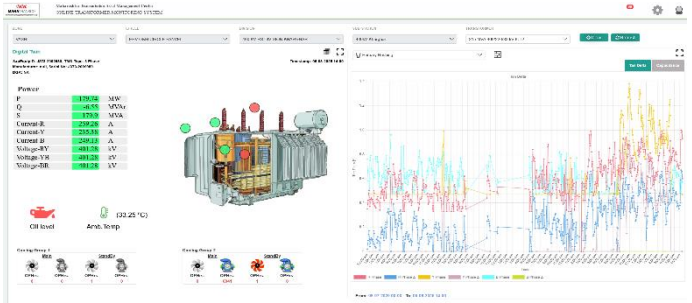
Integration of **8,000+ real-time data points per substation**, enabling comprehensive asset visibility and analytics.

Centralized monitoring and remote access of **561 protection relays**, enhancing system oversight and operational control.



Automated fault analysis and generation of detailed disturbance reports within minutes, significantly accelerating decision-making.

Deployment of **online Dissolved Gas Analysis (DGA) and bushing condition monitoring** for critical power transformers to enable predictive maintenance.



Marked improvement in fault diagnostics, faster root cause identification , Close monitoring and reduced restoration time, strengthening overall grid reliability.

Standardization of Signal list (IEC-61850/ 104) , IP addressing , Relay DR configuration etc.

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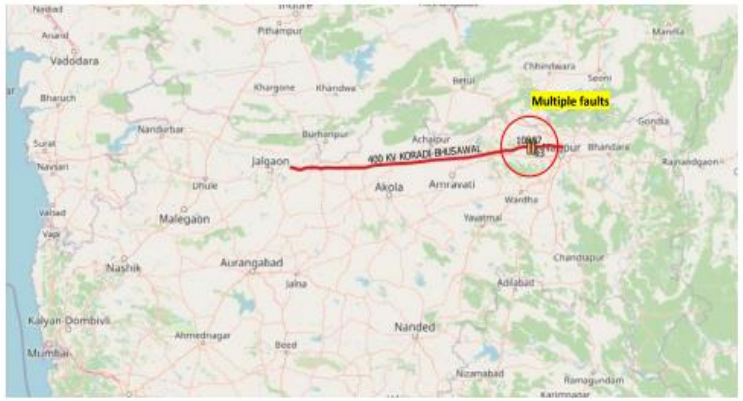
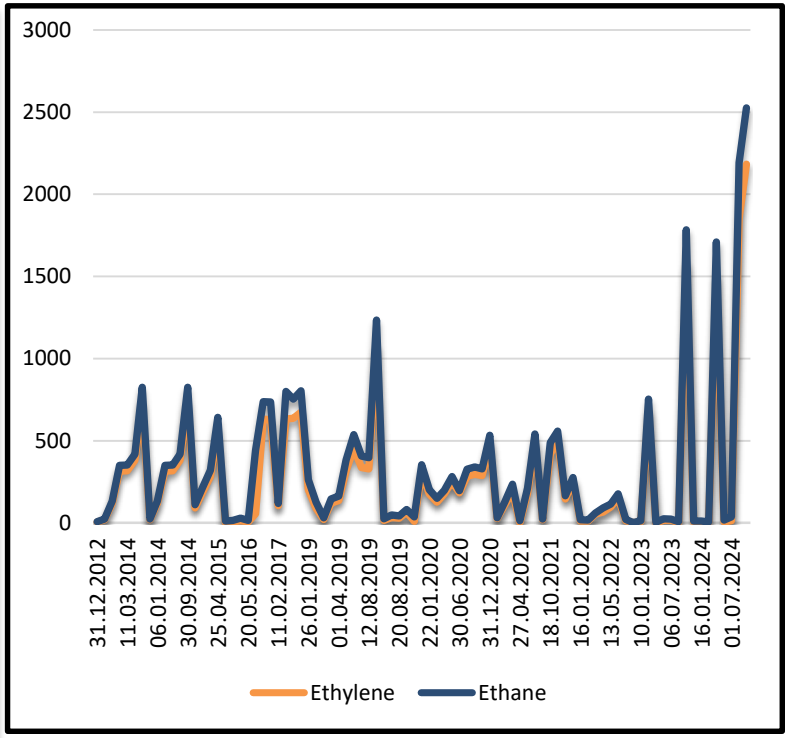
VMS SYSTEM

RAS/AFAS SYSTEM



SR_NO	DATE_OF_FAULT	TIME_OF_FAULT	PH	TOWER_ID	LATITUDE	LONGITUDE	TOW_NUM
1	23-03-2022	5:19	Y	1680	21.2489985	78.6970066	109
2	4/11/2022	1:56	B	1664	21.247588	78.75807	93
3	21-12-2022	18:56	B	1668	21.2480257	78.7424978	97
4	21-12-2022	22:58	B	1668	21.2480257	78.7424978	97

TMS SYSTEM



400 kV Padgha SS ICT 4 B-ph DGA analysis

Conclusion

TRANSFORMATION OF AGING EHV SUBSTATION ASSETS THROUGH AUTOMATION & DIGITAL TECHNOLOGIES FOR IMPROVED SYSTEM RELIABILITY

- MTAMC has demonstrated strong effectiveness in transforming and modernizing aging assets.
- A data-driven, centralized asset management framework has been successfully implemented.
- The initiative is planned for expansion across all MSETCL zones to ensure uniform operational excellence.
- Future roadmap includes integration of AI/ML-driven analytics, digital twin technologies, and enhanced renewable energy integration.
- MTAMC strengthens the foundation for a reliable, secure, and Net-Zero–ready grid.

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THANK YOU

For discussions/suggestions/queries email:

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